
확률 및 랜덤변수

Homework 2

202220953

신현준

#1

$$(a) f_x(x) = \frac{dF_x(x)}{dx} = \begin{cases} 0 & (x < 0) \\ 2e^{-2x} & (x \geq 0) \end{cases}$$

$$(b) P_r(1 < x < 2) = P_r(x < 2) - P_r(x < 1) = F_x(2) - F_x(1) = (1 - e^{-4}) - (1 - e^{-2}) \\ = e^{-2} - e^{-4}$$

$$(c) X \sim \text{Exponential RV} \implies E[X] = \frac{1}{\lambda} = \left(\frac{1}{2}\right)$$

$$\therefore P_r\left(x < \frac{1}{2}\right) = F_x\left(\frac{1}{2}\right) = 1 - e^{-1}$$

#2

(a)

$X=x$ 로 고정되면 $Y=x+N$ 이므로 $Y|X$ 는 가우시안 분포를 따른다. (평균 x , 표준편차 σ_N)

$$f_{X|Y}(x|y) = \frac{f_X(x)f_{Y|X}(y|x)}{f_Y(y)}$$

$$f_X(x) \approx \exp\left(-\frac{(x-\mu_x)^2}{2\sigma_x^2}\right)$$

$$f_{Y|X}(y|x) \approx \exp\left(-\frac{(y-x)^2}{2\sigma_N^2}\right)$$

$$\Rightarrow f_{X|Y}(x|y) \approx \exp\left(-\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-x)^2}{2\sigma_N^2}\right)$$

$$\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-x)^2}{\sigma_N^2}$$

$$= x^2\left(\frac{1}{\sigma_x^2} + \frac{1}{\sigma_N^2}\right) - 2x\left(\frac{\mu_x}{\sigma_x^2} + \frac{y}{\sigma_N^2}\right) + \text{Const}$$

$$= \frac{(x-\mu')^2}{\sigma'^2}$$

$$\frac{1}{\sigma'^2} = \frac{1}{\sigma_x^2} + \frac{1}{\sigma_N^2} \rightarrow \boxed{\sigma'^2 = \frac{\sigma_x^2 \sigma_N^2}{\sigma_x^2 + \sigma_N^2}}$$

$$\begin{aligned} \frac{\mu'}{\sigma'^2} &= \frac{\mu_x}{\sigma_x^2} + \frac{y}{\sigma_N^2} \rightarrow \mu' = \sigma'^2 \left(\frac{\mu_x}{\sigma_x^2} + \frac{y}{\sigma_N^2} \right) \\ &= \boxed{\frac{\sigma_N^2 \mu_x + \sigma_x^2 y}{\sigma_x^2 + \sigma_N^2}} \end{aligned}$$

$$\therefore X|Y=y \sim \mathcal{N}(\mu', \sigma'^2) = \boxed{\mathcal{N}\left(\frac{\sigma_N^2 \mu_x + \sigma_x^2 y}{\sigma_x^2 + \sigma_N^2}, \frac{\sigma_x^2 \sigma_N^2}{\sigma_x^2 + \sigma_N^2}\right)}$$



(b)

$$E[X | Y=y] = \frac{\sigma_u^2 \mu_x + \sigma_x^2 y}{\sigma_x^2 + \sigma_u^2}$$

$$= \boxed{\frac{\sigma_x^2}{\sigma_x^2 + \sigma_u^2} y + \frac{\sigma_u^2}{\sigma_x^2 + \sigma_u^2} \mu_x}$$

(c) σ_u^2 이 점점 커지면, $E[X]$ 에서 y 의 계수는 작아지고, μ_x 의 계수가 커진다.

→ 평균값이 μ_x 에 수렴 ☆

또한 $\sigma' = \frac{\sigma_x^2 \sigma_u^2}{\sigma_x^2 + \sigma_u^2} \rightarrow \underline{\sigma_x^2 \text{으로 수렴}} \star$

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$$\hat{X} = \arg \max_{x \in \{-1, +1\}} P_{X|Y}(x|y) \rightarrow P_{X|Y}(x|y) \text{의 값이 가장 큰 } x = \hat{X}$$

(a)

$$P_{X|Y}(x|y) = \frac{p_x(x) f_{Y|X}(y|x)}{f_Y(y)} \quad \text{이므로 성립}$$

$$(b) \quad \hat{X} = \begin{cases} +1 & (p_x(+1) f_{Y|X}(y|+1) > p_x(-1) f_{Y|X}(y|-1)) \\ -1 & (p_x(+1) f_{Y|X}(y|+1) < p_x(-1) f_{Y|X}(y|-1)) \end{cases}$$

(c)

$$Y|X=x \sim N(x, \frac{\sigma^2}{2}) \text{ or } \text{IID}$$

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi} \cdot \frac{\sigma}{\sqrt{2}}} \exp\left(-\frac{(y-x)^2}{2 \cdot \frac{\sigma^2}{2}}\right) = \frac{1}{\sqrt{\pi} \sigma} \exp\left(-\frac{(y-x)^2}{\sigma^2}\right)$$

$$p_x(1) f_{Y|X}(y|1) = \frac{1}{a} \cdot \frac{1}{\sqrt{\pi} \sigma} \exp\left(-\frac{(y-1)^2}{\sigma^2}\right)$$

$$p_x(-1) f_{Y|X}(y|-1) = \left(1 - \frac{1}{a}\right) \frac{1}{\sqrt{\pi} \sigma} \exp\left(-\frac{(y+1)^2}{\sigma^2}\right)$$

$$p_x(1) f_{Y|X}(y|1) > p_x(-1) f_{Y|X}(y|-1) \text{ if } \text{for}$$

$$\exp\left(-\frac{(y-1)^2}{\sigma^2}\right) > (a-1) \exp\left(-\frac{(y+1)^2}{\sigma^2}\right)$$

$$-\frac{(y-1)^2}{\sigma^2} > \ln(a-1) - \frac{(y+1)^2}{\sigma^2}$$

$$-(y-1)^2 > \sigma^2 \ln(a-1) - (y+1)^2$$

$$(y+1)^2 - (y-1)^2 > \sigma^2 \ln(a-1)$$

$$4y > \sigma^2 \ln(a-1)$$

$$\boxed{y > \frac{\sigma^2}{4} \ln(a-1)} \quad \text{or} \quad \hat{X} = +1$$

$$\therefore b = \frac{\sigma^2}{4} \ln(a-1) \quad \star$$

#4

(a)

$$\begin{aligned}
 \int_0^1 \int_x^2 c(x+y) dy dx &= \int_0^1 cx(x-x^2) + \frac{1}{2}c(x^2-x^4) dx \\
 &= \int_0^1 \left(cx^2 - cx^3 + \frac{1}{2}cx^2 - \frac{1}{2}cx^4 \right) dx \\
 &= \int_0^1 \left(-\frac{1}{2}cx^4 - cx^3 + \frac{3}{2}cx^2 \right) dx \\
 &= \left. -\frac{1}{10}cx^5 - \frac{1}{4}cx^4 + \frac{1}{2}cx^3 \right|_0^1 = -\frac{1}{10}c - \frac{1}{4}c + \frac{1}{2}c = 1
 \end{aligned}$$

$$-2c - 5c + 10c = 20$$

$$3c = 20$$

$$\therefore c = \frac{20}{3}$$

(b)

$$\begin{aligned} f_x(x) &= \int_{-\infty}^{\infty} f_{x|Y}(x,y) dy \\ &= \int_x^1 \frac{20}{3}(x+y) dy = \frac{20}{3}x(x-x^2) + \frac{10}{3}(x^2-x^4) \\ &= \frac{10}{3}(2x^2-2x^3+x^2-x^4) = \frac{10}{3}x^2(3-2x-x^2) \\ &= \frac{10}{3}x^2(1-x)(3+x) \quad (0 \leq x \leq 1) \end{aligned}$$

$$\therefore f_x(x) = \begin{cases} \frac{10}{3}x^2(1-x)(3+x) & (0 \leq x \leq 1) \\ 0 & (\text{otherwise}) \end{cases}$$



(c)

y 가 고정일 때 x 의 범위 $x^2 \leq y \leq x \rightarrow y \leq x \leq \sqrt{y}$

$$\begin{aligned} f_Y(y) &= \int_y^{\sqrt{y}} \frac{20}{3} (x+y) dx = \frac{10}{3} (y - y^2) + \frac{20}{3} y (\sqrt{y} - y) \\ &= \frac{10}{3} y (1 - y + 2\sqrt{y} - 2y) = \frac{10}{3} y (1 + 2\sqrt{y} - 3y) \end{aligned}$$

$$\therefore f_Y(y) = \begin{cases} \frac{10}{3} y (1 + 2\sqrt{y} - 3y) & (0 \leq y \leq 1) \\ 0 & (\text{otherwise}) \end{cases}$$



(d)

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\frac{20}{3}(x+y)}{\frac{10}{3}x^2(1-x)(3+x)}$$

$$f_{Y|X}(y|x) = \frac{2(x+y)}{x^2(1-x)(3+x)} \quad (x^2 \leq y \leq x, 0 \leq x \leq 1)$$

$$\begin{aligned} F_{Y|X}(y|x) &= \int_{-\infty}^y f_{Y|X}(y'|x) dy' \\ &= \int_{x^2}^y \frac{2(x+y')}{x^2(1-x)(3+x)} dy' = \frac{2x(y-x^2) + (y^2-x^4)}{x^2(1-x)(3+x)} \end{aligned}$$

$$F_{Y|X}(y|x) = \begin{cases} 0 & (y < x^2) \\ \frac{2xy + y^2 - 2x^3 - x^4}{x^2(1-x)(3+x)} & (x^2 \leq y < x) \\ 1 & (y \geq x) \end{cases}$$

$$\begin{aligned}
 E[Y | X=x] &= \int_{-\infty}^{\infty} y f_{Y|X}(y|x) dy \\
 &= \int_x^1 y \cdot \frac{2(1+y)}{x^2(1-x)(3+x)} dy = \frac{x(x^3-x^4) + \frac{2}{3}(x^3-x^6)}{x^2(1-x)(3+x)} \\
 &= \frac{3x^3-3x^5+2x^3-2x^6}{3x^2(1-x)(3+x)} = \frac{x(5-3x^2-2x^3)}{3(1-x)(3+x)} \\
 &= \frac{x(2x^2+5x+5)}{3(3+x)}
 \end{aligned}$$

$$\begin{aligned}
 E[Y^2 | X=x] &= \int_x^1 y^2 \frac{2(1+y)}{x^2(1-x)(3+x)} dy = \frac{\frac{2}{3}x(x^3-x^6) + \frac{1}{4}(x^4-x^9)}{x^2(1-x)(3+x)} \\
 &= \frac{x^2(3x^3+7x^2+7x+7)}{6(1+x)}
 \end{aligned}$$

$$V[Y|X=x] = \frac{x^2(2x^3 + 7x^2 + 7x + 7)}{6(x+3)} - \left\{ \frac{x(2x^2 + 5x + 5)}{3(3+x)} \right\}^2$$

$$= \frac{2x^2(2x^3 + 7x^2 + 7x + 7)(x+3) - 2x^2(2x^2 + 5x + 5)^2}{(6(x+3))^2}$$

$$= \frac{x^2(1-x)^2(x^2 + 10x + 13)}{(6(x+3))^2}$$

#5

$$\begin{aligned} (a) \quad p_{X|Y}(x'|y) &= \frac{p_{X,Y}(x'|y)}{f_Y(y)} = \frac{\cancel{p_X(i)} f_{Y|X}(y|x')}{\sum_i \cancel{p_X(i)} f_{Y|X}(y|i)} \\ &= \frac{f_{Y|X}(y|x')}{e^2 + e + e^{-1}} \end{aligned}$$

$$p_{X|Y}(1|y) = \frac{e^2}{e^2 + e + e^{-1}}$$

$$p_{X|Y}(2|y) = \frac{e}{e^2 + e + e^{-1}}$$

$$p_{X|Y}(3|y) = \frac{e^{-1}}{e^2 + e + e^{-1}}$$

(b)

$$P_t(x=i | Y=y) = P_{X|Y}(i|y) = \frac{f_{Y|X}(y|i)}{\sum_k f_{Y|X}(y|k)} = \frac{e^{s_i}}{\sum_{j=1}^3 e^{s_j}}$$

$$e^{s_i} = f_{Y|X}(y|i)$$

$$s_i = \ln(f_{Y|X}(y|i)) \rightarrow \begin{cases} s_1 = \ln(e^2) = 2 \\ s_2 = \ln(e^1) = 1 \\ s_3 = \ln(e^{-1}) = -1 \end{cases}$$

(c) $s_1 = 2, s_2 = 1, s_3 = -1$

입력의 싹 값 s_i 에 $\exp(\cdot)$ 를 취하여 e^{s_i} 로 양속화 + 합으로 나눠 정규화

⇒ 항상 양수이고 합이 1인 확률분포로 변환하셈

⊕ 가장 큰 s_i 에 가장 큰 확률을 주지만, 부드럽게 분배한다는 점에서 hardmax 의 'soft' 버전.

∴ 간단히 Bayes의 식이 softmax 형태로 표현된다.

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(a)

$$\frac{d}{dt} e^{tX} = X e^{tX} \rightarrow \frac{d^n}{dt^n} e^{tX} = X^n e^{tX}$$

$$\rightarrow \frac{d^n}{dt^n} E[e^{tX}] = E[X^n e^{tX}]$$

$$\rightarrow \frac{d^n}{dt^n} M_X(t) = E[X^n e^{tX}]$$

($t=0$)₂ $\frac{0}{2}$ $\frac{0}{2}$ $\frac{0}{2}$

$$E[X^n] = \frac{d^n}{dt^n} M_X(t)$$

(b)

$$M'_x(t) = (3+4t)e^{3t+2t^2} \rightarrow E[X] = M'_x(0) = 3e^0 = 3$$

$$M''_x(t) = 4e^{3t+2t^2} + (3+4t)^2 e^{3t+2t^2} \rightarrow E[X^2] = M''_x(0) = 4e^0 + 3^2 e^0 = 13$$

$$\therefore V[X] = E[X^2] - \{E[X]\}^2 = 13 - 9 = 4$$

(c) $X \sim N(0,1)$ 01022 $f_x(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$

$$E[e^{itX}] = \int_{-\infty}^{\infty} e^{itX} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left(itX - \frac{x^2}{2}\right)} dx$$

$$= \frac{e^{-\frac{t^2}{2}}}{\sqrt{2\pi}} \underbrace{\int_{-\infty}^{\infty} e^{\frac{-(x-it)^2}{2}} dt}_{2\pi} = e^{-\frac{t^2}{2}}$$

$$-\frac{1}{2} \{ (x-it)^2 - (it)^2 \}$$

2π

(c)

	MGF $M_X(t) = E[e^{tX}]$	CF $\phi_X(t) = E[e^{itX}]$
정의	$t=0$ 근방에서만 e^{tX} 가 유한해야 함	모든 $t \in \mathbb{R}$, e^{itX} 가 유한해야 함
분포 유일 결정	MGF가 $t=0$ 의 근방에서 유일하면, 그 MGF가 분포를 유일하게 결정	$\phi_X = \phi_Y$ 이면 $X=Y$
계산 편의성	실수 변수만 존재 $\rightarrow$ 미분으로 모멘트 추출이 상대적으로 편함.	복소 변수 $\rightarrow$ Fourier 해석 필요.
모멘트 추출	$E[X^n] = M_X^{(n)}(0)$ if exists.	$E[X^n] = (-i)^n \phi_X^{(n)}(0)$ if exists.

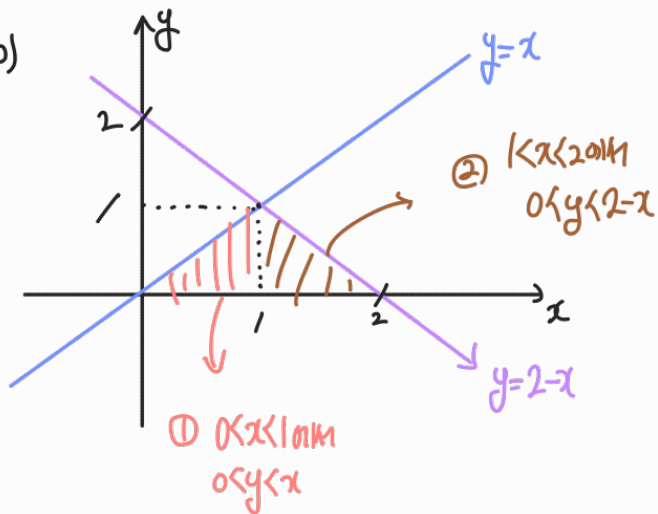
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$$\begin{aligned} (a) \quad \int_0^{\infty} \int_0^1 c(x+y)e^{-x} dy dx &= \int_0^{\infty} \left(cxe^{-x} \cdot 1 + ce^{-x} \cdot \frac{1}{2}x^2 \right) dx \\ &= \int_0^{\infty} \left(\frac{3}{2}cx^2e^{-x} \right) dx \\ &= \frac{3}{2}c \left(\int_0^{\infty} x^2e^{-x} dx + \int_0^{\infty} 2xe^{-x} dx \right) \\ &= \frac{3}{2}c \left(\int_0^{\infty} 2xe^{-x} dx + \int_0^{\infty} 2e^{-x} dx \right) \\ &= 3c = 1 \end{aligned}$$

$\therefore C = \frac{1}{3}$

★

(b)



$$\begin{aligned}
 \textcircled{1} \quad \int_0^1 \int_0^x \frac{1}{2}(x+y)e^{-x} dy dx &= \int_0^1 e^{-x} \cdot \frac{1}{2}x^2 dx = \frac{1}{2} \left\{ [x^2 e^{-x}]_0^1 + \int_0^1 2x e^{-x} dx \right\} \\
 &= \frac{1}{2} \left\{ (-e^{-1}) + [-2x e^{-x}]_0^1 + \int_0^1 2e^{-x} dx \right\} \\
 &= \frac{1}{2} \left\{ (-e^{-1}) + (-2e^{-1}) + [-2e^{-x}]_0^1 \right\} \\
 &= \frac{1}{2} \left\{ -3e^{-1} - 2(e^{-1} - 1) \right\} = \frac{1}{2} (2 - 5e^{-1})
 \end{aligned}$$

$$\textcircled{2} \int_1^2 \int_0^{2-x} \frac{1}{3}(x+y)e^{-x} dy dx$$

$$= \frac{1}{3} \int_1^2 e^{-x} \cdot \left\{ x(2-x) + \frac{1}{2}(2-x)^2 \right\} dx$$

$$= \frac{1}{3} \int_1^2 e^{-x} \cdot \frac{4-x^2}{2} dx = \frac{1}{3} \left[\int_1^2 2e^{-x} dx - \frac{1}{2} \int_1^2 x^2 e^{-x} dx \right]$$

$$= \frac{1}{3} \left[[-2e^{-x}]_1^2 - \frac{1}{2} \left\{ -x^2 e^{-x} \Big|_1^2 + \int_1^2 2x \cdot e^{-x} dx \right\} \right]$$

$$= \frac{1}{3} \left[-2(e^{-2} - e^{-1}) - \frac{1}{2} \left\{ (-4e^{-2} + e^{-1}) + [-2xe^{-x}]_1^2 + \int_1^2 2e^{-x} dx \right\} \right]$$

$$= \frac{1}{3} \left[-2(e^{-2} - e^{-1}) - \frac{1}{2} \left\{ -4e^{-2} + e^{-1} - 4e^{-2} + 2e^{-1} + [-2e^{-x}]_1^2 \right\} \right]$$

$$= \frac{1}{3} \left[-2(e^{-2} - e^{-1}) - \frac{1}{2} \left\{ -8e^{-2} + 3e^{-1} - 2e^{-2} + 2e^{-1} \right\} \right]$$

$$= \frac{1}{3} \left\{ -2e^{-2} + 2e^{-1} - \frac{1}{2}(-10e^{-2} + 5e^{-1}) \right\}$$

$$= \boxed{\frac{1}{3} \left(3e^{-2} - \frac{1}{2}e^{-1} \right)}$$

$$\rightarrow P(A) = \textcircled{1} + \textcircled{2} = \frac{1}{2}(2 - 5e^{-1}) + \frac{1}{3}(3e^{-2} - \frac{1}{2}e^{-1})$$

$$= 1 - \frac{5}{2}e^{-1} + e^{-2} - \frac{1}{6}e^{-1} = \boxed{1 - \frac{8}{3}e^{-1} + e^{-2}}$$

$$\therefore f_{X,Y|A}(x,y) \begin{cases} \frac{f_{X,Y}(x,y)}{P(A)} = \frac{\frac{1}{3}(x+y)e^{-x}}{1 - \frac{8}{3}e^{-1} + e^{-2}} = \boxed{\frac{e^2(x+y)e^{-x}}{3e^2 - 8e + 3}} \\ 0 \text{ (otherwise)} \end{cases}$$

$$(0 \leq y \leq x, x+y \leq 2)$$

(c)

$$f_{x|A}(x) = \int f_{x,y|A}(x,y) dy$$

$$\textcircled{1} 0 \leq x \leq 1 \quad (0 \leq y \leq x)$$

$$\int_0^x \frac{e^2(x+y)e^{-x}}{3e^2 - 8e + 3} dy = \frac{e^2 e^{-x}}{3e^2 - 8e + 3} \cdot \left(x^2 + \frac{1}{2}x^2\right) = \frac{e^2 e^{-x}}{3e^2 - 8e + 3} \cdot \frac{3}{2}x^2$$

$$\textcircled{2} 1 \leq x \leq 2 \quad (0 \leq y \leq 2-x)$$

$$\int_0^{2-x} \frac{e^2 e^{-x}}{3e^2 - 8e + 3} dy = \frac{e^2 e^{-x}}{3e^2 - 8e + 3} \left\{ x(2-x) + \frac{1}{2}(2-x)^2 \right\} = \frac{e^2 e^{-x}}{3e^2 - 8e + 3} \cdot \frac{4-x^2}{2}$$

$$\therefore f_{x|A}(x) = \begin{cases} \frac{3x^2 e^{2-x}}{2(3e^2 - 8e + 3)} & (0 \leq x < 1) \\ \frac{e^{2-x}(4-x^2)}{2(3e^2 - 8e + 3)} & (1 \leq x \leq 2) \end{cases}, \quad (\text{otherwise } 0)$$

(d)

$$f_{x,y|A}(x,y) = \frac{(x+y)e^{-x}}{3P(A)} \text{ if } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 2-x,$$

$$f_{x|A}(x) = \begin{cases} \frac{x^2 e^{-x}}{2P(A)} & (0 \leq x \leq 1) \\ \frac{(4-x^2)e^{-x}}{6P(A)} & (1 \leq x \leq 2) \\ 0 & (\text{otherwise}) \end{cases}$$

$$\begin{aligned} E[X|A] &= \int_0^1 x \cdot \frac{x^2 e^{-x}}{2P(A)} dx + \int_1^2 x \cdot \frac{(4-x^2)e^{-x}}{6P(A)} dx \\ &= \frac{1}{2P(A)} \int_0^1 x^3 e^{-x} dx + \frac{1}{6P(A)} \int_1^2 (4x-x^3)e^{-x} dx \\ &\quad \downarrow \qquad \qquad \qquad \downarrow \\ &\quad 6 - \frac{16}{e} \qquad \qquad \qquad -\frac{8}{e} + \frac{26}{e^2} \end{aligned}$$

$$= \frac{3(6 - \frac{16}{e}) + (-\frac{8}{e} + \frac{26}{e^2})}{6P(A)} = \frac{18 - \frac{56}{e} + \frac{26}{e^2}}{6P(A)}$$

$$\Rightarrow E[X|A] = \frac{9e^2 - 28e + 13}{3e^2 - 8e + 3} \approx 0.991$$

$$E[X^2|A] = \frac{1}{2P(A)} \int_0^1 x^4 e^{-x} dx + \frac{1}{6P(A)} \int_1^2 x^2(4-x^2)e^{-x} dx$$

$\xrightarrow{24 - \frac{65}{e}}$
 $\xrightarrow{-\frac{45}{e} + \frac{129}{e^2}}$

$$= \frac{3(24 - \frac{65}{e}) + (-\frac{45}{e} + \frac{129}{e^2})}{6P(A)} = \frac{4(9e^2 - 30e + 16)}{3e^2 - 8e + 3} \approx 1.113$$

$$\therefore V[X|A] = E[X^2|A] - \{E[X|A]\}^2 = 1.113 - (0.991)^2 = 0.131$$

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$$(a) P(X \leq 85) \rightarrow Z = \frac{X - \mu_0}{\sigma_0} \leq \frac{85 - 70}{10} = 1.5$$

$$\therefore P(Z \leq 1.5) = \boxed{1 - Q(1.5)}$$

$$(b) P(60 \leq X \leq 80) \rightarrow \frac{60 - 70}{10} \leq Z \leq \frac{80 - 70}{10} \rightarrow -1 \leq Z \leq 1$$

$$P(-1 \leq Z \leq 1) = \Phi(1) - \Phi(-1) = 2\Phi(1) - 1 = 1 + \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) - 1 = \boxed{\operatorname{erf}\left(\frac{1}{\sqrt{2}}\right)}$$

$$(c) P(X > 90) \rightarrow Z > \frac{90 - 70}{10} = 2 \rightarrow Z > 2$$

$$P(Z > 2) = 1 - \Phi(2) = 1 - \frac{1}{2} (1 + \operatorname{erf}(\sqrt{2})) = \frac{1}{2} (1 - \operatorname{erf}(\sqrt{2})) = \boxed{\frac{1}{2} \operatorname{erfc}(\sqrt{2})}$$

#9

$$(a) \quad f_h(h) = \begin{cases} e^{-h} & (h \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$

$$\begin{aligned} (b) \quad P(\text{error}) &= \sum_b P(B=b) P(\text{error} | B=b) && \text{B와 H는 독립} \\ &= \sum_b P(B=b) \int_{-\infty}^{\infty} P(\text{error} | B=b, H=h) \underline{f_{h|b}(h|b)} \, dh \\ &= \sum_b P(B=b) \int_{-\infty}^{\infty} P(\text{error} | B=b, H=h) f_h(h) \, dh \\ &= \underbrace{\sum_b P(B=b)}_{\frac{1}{2} + \frac{1}{2} = 1} \underbrace{\int_{-\infty}^{\infty} Q(\sqrt{2\rho}h) f_h(h) \, dh}_{\text{b와 무관}} = \int_0^{\infty} Q(\sqrt{2\rho}h) e^{-h} \, dh \end{aligned}$$

$$0/1 \text{ } Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-\frac{t^2}{2}} dt \quad 0/03$$

$$Q(\sqrt{2\rho h}) = \frac{1}{\sqrt{2\pi}} \int_{\sqrt{2\rho h}}^{\infty} e^{-\frac{t^2}{2}} dt$$

$$P(\text{error}) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} \left(\int_{\sqrt{2\rho h}}^{\infty} e^{-\frac{t^2}{2}} dt \right) \cdot e^{-h} dh$$

$$0 \leq h \leq \infty$$

$$\sqrt{2\rho h} \leq t \leq \infty \rightarrow 2\rho h \leq t^2 \rightarrow h \leq \frac{t^2}{2\rho}$$

$$P(\text{error}) = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} \int_0^{\frac{t^2}{2\rho}} e^{-\frac{t^2}{2}} e^{-h} dh dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{t^2}{2}} \left(\int_0^{\frac{t^2}{2\rho}} e^{-h} dh \right) dt$$

$1 - e^{-\frac{t^2}{2\rho}}$

$$= \boxed{\frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{t^2}{2}} dt} + \boxed{\frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{t^2}{2} - \frac{t^2}{2e}} dt}$$

$= \frac{1}{2}$

↘

$$\frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{t^2}{2} \cdot \frac{e+1}{e}} dt$$

$$\sigma^2 = \frac{e}{e+1} \text{ की जगह}$$

$$\sigma \cdot \frac{1}{\sigma\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{t^2}{2\sigma^2}} dt = \frac{1}{2}\sigma$$

$= \frac{1}{2}$

$= \left(\frac{1}{2} \sqrt{\frac{e}{e+1}} \right)$

$$= \frac{1}{2} + \frac{1}{2} \sqrt{\frac{e}{e+1}}$$

$$= \boxed{\frac{1}{2} \left(1 + \sqrt{\frac{e}{e+1}} \right)}$$

☆☆

10

(a)

$$E[X|Z=z] = \int_{-\infty}^{\infty} x \cdot f_{X|Z}(x|z) dx = \int_0^z x \cdot \frac{1}{z} dx = \frac{1}{z} \cdot \frac{1}{2} z^2 = \frac{1}{2} z$$

(b)

$$E[X] = \int_{-\infty}^{\infty} f_Z(z) E[X|Z=z] dz = \int_0^1 1 \cdot \frac{1}{2} z dz = \frac{1}{4}$$

(c)

$$E[X^2|Z=z] = \int_0^z x^2 \cdot \frac{1}{z} dx = \frac{1}{z} \cdot \frac{1}{3} z^3 = \frac{1}{3} z^2$$

(d)

$$E[X^2] = \int_{-\infty}^{\infty} f_z(z) E[X^2 | Z=z] dz = \int_0^1 1 \cdot \frac{1}{3} z^2 dz = \frac{1}{9}$$

(e)

$$\text{Var}[X] = E[X^2] - (E[X])^2 = \frac{1}{9} - \left(\frac{1}{4}\right)^2 = \frac{7}{144}$$

#11

$$(a) \iiint f_{x,y,z}(x,y,z) dx dy dz$$

$$= \int_0^{\infty} \int_0^z \int_0^z c e^{-z} dx dy dz = c \int_0^{\infty} z^2 e^{-z} dz$$

$$= 2c = 1 \rightarrow c = \frac{1}{2}$$

$$(b) f_z(z) = \iint f_{x,y,z}(x,y,z) dx dy$$

$$= \int_0^z \int_0^z \frac{1}{2} e^{-z} dx dy = \frac{1}{2} z^2 e^{-z}$$

$$f_{x,y|z}(x,y|z) = \frac{f_{x,y,z}(x,y,z)}{f_z(z)} = \frac{\frac{1}{2}e^{-z}}{\frac{1}{2}z^2e^{-z}} = \frac{1}{z^2}$$

$$\therefore f_{x,y|z}(x,y|z) = \begin{cases} \frac{1}{z^2} & (0 \leq x \leq z, 0 \leq y \leq z) \\ 0 & (\text{otherwise}) \end{cases}$$



(c)

$$f_{x|z}(x|z) = \int f_{x,y|z}(x,y|z) dy = \int_0^z \frac{1}{z^2} dy = \frac{1}{z}$$

같은 방법으로 $f_{y|z}(y|z) = \frac{1}{z} \Rightarrow f_{x|z}(x|z) f_{y|z}(y|z) = f_{x,y|z}(x,y|z)$

이므로 조건부 독립임 ★ (자세한것도 일치) $\left(\frac{1}{z^2}\right)$

#12

$$f_z(z) = \lambda e^{-\lambda z} \quad (z \geq 0, \text{ otherwise } 0)$$

$$f_{x|z}(x|z) = z e^{-zx} \quad (x \geq 0, \text{ otherwise } 0)$$

$$f_{y|z}(y|z) = z e^{-zy} \quad (y \geq 0, \text{ otherwise } 0)$$

$z = z$ 가 주어지면 X, Y 가 독립이므로

$$\begin{aligned} f_{X,Y|z}(x,y|z) &= f_{x|z}(x|z) \cdot f_{y|z}(y|z) \\ &= z^2 e^{-\lambda(x+y)} \quad (x, y \geq 0, \text{ otherwise } 0) \end{aligned}$$

$$f_{X,Y}(x,y) = \int_{-\infty}^{\infty} f_z(z) f_{X,Y|z}(x,y|z) dz$$

$$= \int_0^{\infty} \lambda e^{-\lambda z} \cdot z^2 e^{-\lambda(x+y)} dz$$

$$= \lambda \int_0^{\infty} z^2 e^{-\lambda(x+y+z)} dz = \frac{2\lambda}{(\lambda+x+y)^3} \quad (x, y \geq 0)$$

$$\therefore f_{x,y}(x,y) = \begin{cases} \frac{2\lambda}{(\lambda+x+y)^3} & (x,y \geq 0) \\ 0 & (\text{otherwise}) \end{cases}$$



$$(b) \quad f_x(x) = \int_{-\infty}^{\infty} f_{x,y}(x,y) dy = \int_0^{\infty} \frac{2\lambda}{(\lambda+x+y)^3} dy$$

$$u = \lambda + x + y \rightarrow \begin{cases} du = dy & \text{if } 0 \leq y < \infty \\ \lambda + x \leq u < \infty \end{cases} \quad f_x(x) = \int_{\lambda+x}^{\infty} \frac{2\lambda}{u^3} du = \frac{\lambda}{(\lambda+x)^2}$$

$$\Rightarrow f_x(x) = \frac{\lambda}{(\lambda+x)^2}, \quad f_y(y) = \frac{\lambda}{(\lambda+y)^2}$$

$$f_{X,Y}(x,y) \neq f_x(x)f_y(y) \text{ 이므로 } \underline{X, Y \text{ 는 독립이 아니다}} \quad \triangle$$